

LAB2SIM: FROM WET-LAB EVIDENCE TO ROBOT-READY USD SCENES

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ABSTRACT

A wet-lab protocol names the instruments a robot must touch. A physics simulator needs USD: articulated joints, colliders, materials, and a floor the robot can stand on. Lab2Sim is an agent that compiles the first document into the second. Given a text instruction, a catalog photograph, or a demonstration video, the agent searches out the real instrument, writes an asset specification that fixes its task-critical geometry and articulation, iterates articulated USD with a Blender skill, and admits the result through ConvertAsset into an Isaac scene a Lift2 can enter. We show the laboratories the agent produces—Fehling baths, titration stands, powder weighing, a drying oven, and stopper, glass-rod, stir-bar, and water-bath stations—and trace two instruments, the IKA OVEN 125 and a traditional burette, from manufacturer evidence through the agent-written specification to the admitted USD.

1 INTRODUCTION

“Open the IKA OVEN 125 and unload the glassware” is a legal line in a lab notebook. It is not a scene. Before a robot can rehearse it, the oven needs a door on a hinge, a handle a gripper can close on, shelves that count as placement surfaces, and a laboratory floor for a Lift2 to stand on. The protocol names the instrument; the simulator needs its USD.

Two lines of work approach that gap from opposite sides. Real-to-sim systems reconstruct a workspace that already exists, from a scan or a photograph (Torné Villasevil et al., 2024; Dai et al., 2024; Li et al., 2024; Chen et al., 2024). Generative scene agents furnish rooms and write task programs from a prompt (Yang et al., 2024; Wang et al., 2024a;b). A wet-lab station starts from neither. Its instruments are specific products—a burette with a working stopcock, an oven whose door travels 180°—named by a protocol rather than seen in a scan. Laboratory robotics has policies and simulators, but no compiler between the notebook and the scene.

Lab2Sim is that compiler, run as an agent (Figure 1). Text, image, and video enter as one request. The agent searches for the real device and keeps its manufacturer evidence; writes an asset specification that fixes envelope, joints, grasp regions, and placement surfaces; iterates a USD candidate with a Blender skill until it matches; and hands it to ConvertAsset, whose admission places it in an Isaac laboratory (NVIDIA, 2024). The specification is the agent’s central artifact: everything upstream is evidence for it, everything downstream is checked against it.

We make three claims. The compiler runs end to end on real instruments. Its laboratories span different rooms and manipulation verbs (Figure 2). And the task-critical features of an instrument survive from evidence to specification to admitted USD, which we show on the OVEN 125 and a traditional burette (Figure 3).

2 RELATED WORK

Real-to-sim reconstruction. RialTo scans a scene into a digital twin for policy robustification (Torné Villasevil et al., 2024). Digital cousins build interactive lookalikes from a single RGB image (Dai et al., 2024). SIMPLER matches real robot setups so that simulated evaluation tracks real evaluation (Li et al., 2024). URDFormer predicts articulated URDF scenes directly from real photographs (Chen et al., 2024). Lab2Sim shares the goal of an interactive twin but takes a different

input: a protocol and the manufacturer’s own evidence for the named instrument, rather than a scan or photograph of a workspace that already exists.

Generative scene agents. GenSim writes simulation task code from language (Wang et al., 2024a). RoboGen proposes tasks, scenes, and supervision in a propose–generate–learn loop (Wang et al., 2024b). Holodeck lays out 3D rooms from a prompt using Objaverse meshes (Yang et al., 2024). Image-conditioned 3D generators such as Hunyuan3D propose textured geometry from a single view (Tencent Hunyuan3D, 2025). Lab2Sim calls generators of this kind as tools inside its Blender skill. What it adds is the layer these agents leave implicit: a written specification of the instrument’s joints and contact surfaces, and an admission step that decides whether the generated USD may enter the robot’s laboratory.

3 LAB2SIM

Lab2Sim treats a protocol as a compilation request and moves it through five artifacts: a *request*, the *evidence* for each instrument, an *asset specification*, a *USD candidate*, and an *admitted scene*. Each stage consumes the previous artifact; once the specification exists, every later stage is checked against it.

Understanding the request. A request arrives as text, a photograph, or a demonstration video. The agent reads out which instruments the experiment needs, which must articulate, and what the robot will do to each. “Unload the oven” yields a hinged door, a graspable handle, and shelf placement; a catalog photo of the OVEN 125 yields the same demands with the model already identified; a demo video adds the order in which devices are touched. The three modalities converge on one table of asset requests.

Searching for the real instrument. For every request row the agent finds the actual product. Manufacturer pages, datasheets, and official photographs are hashed into an evidence register and graded as exact-model, family, derived, or unknown. Identity is a model number, not “a lab oven”. This is what stops the specification from inventing a hinge the instrument does not have, and what lets Figure 3 put a catalog photograph beside the USD.

Writing the asset specification. From the evidence the agent fixes the envelope, the decomposition into links, the prim tree, joint types and limits, grasp regions, placement surfaces, state variables, and a QA matrix. The OVEN 125 specification fixes a $700 \times 825 \times 650$ mm body, a revolute door hinge with $0\text{--}180^\circ$ travel, a handle collider on the door, and six shelves as placement surfaces. The burette

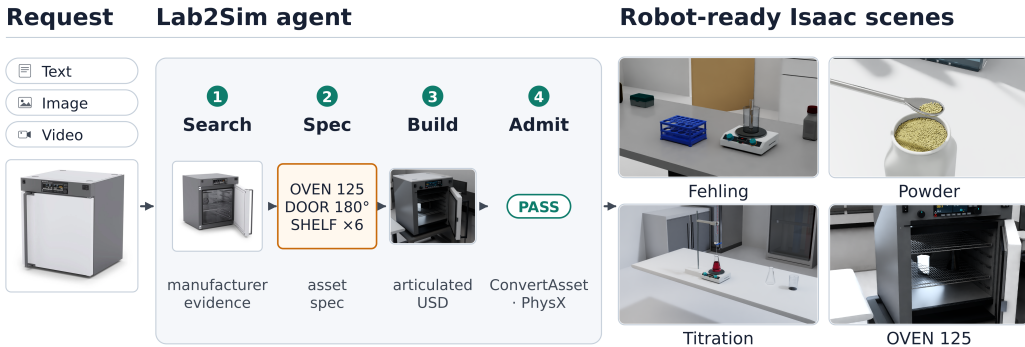


Figure 1: Text, image, and video form one request interface. Lab2Sim searches manufacturer evidence, writes an asset specification, builds articulated USD with a Blender skill (Blender Foundation, 2024), and admits it through ConvertAsset. Hunyuan3D is an optional geometry frontend (Tencent Hunyuan3D, 2025). Right: Fehling, powder, titration, and OVEN 125 Isaac scenes the agent delivered from independent requests.

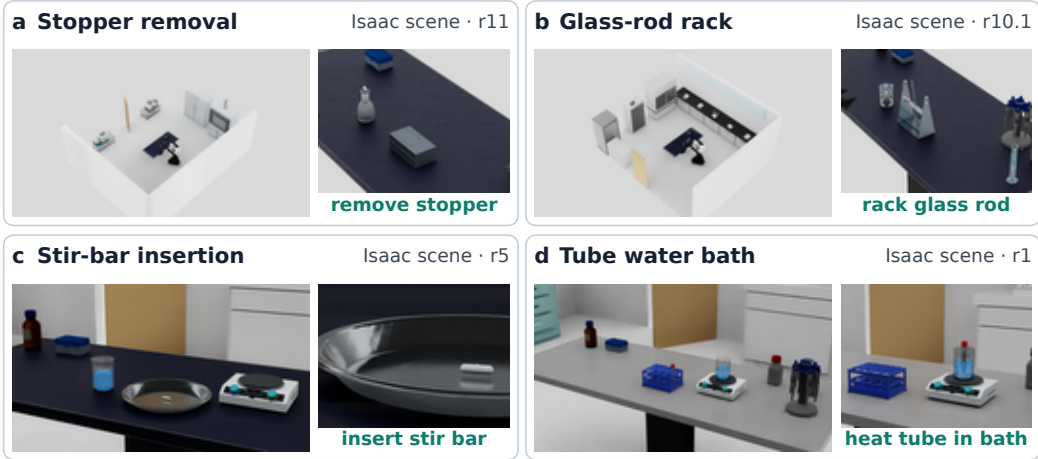


Figure 2: Four Lab2Sim laboratories beyond those in Figure 1. Each panel shows the delivered Isaac station and, inset, the objects the task acts on. Top: two eBench rooms, shown as cutaways with a Lift2 at the bench, for removing a ground-glass stopper (teaching room, r11) and racking a glass rod (analytical-instrumentation room, r10.1). Bottom: two bench-level liquid stations, for dropping a stir bar into a beaker (r5) and heating a tube in a water bath (r1).

specification fixes a 560 mm, $\varnothing 12$ mm tube, a revolute stopcock with a 0–90° target, a grasp region on the handle, and a mount frame the stand receives through a scene-level fixed joint. Fields the evidence does not support stay open. Generation may rebuild geometry; it may not change these numbers or rename these contracts.

Generating the USD candidate. A skill package turns the specification into articulated USD. Procedural modelling in Blender is the always-on route (Blender Foundation, 2024); Hunyuan3D is an optional geometry frontend the same skill calls when it is reachable (Tencent Hunyuan3D, 2025). The agent iterates until a static articulation check under Isaac Kit Python matches the specification: joint names, axes, limits, and colliders. A failing candidate returns to generation with the mismatch named.

Admitting the scene. ConvertAsset is the existing conversion and physics-qualification compiler for meshes, materials, and PhysX fitness; Lab2Sim hands it the candidate and does not reimplement it. A candidate that fails conversion is an explicit gap. One that passes receives a promotion receipt, is bound into a scenario, compiled, and spawned in Isaac Sim with a Lift2. Admission is the moment the protocol becomes a place the robot can stand.

4 COMPILED LABORATORIES

Two questions decide whether the compiler is useful. Which laboratories can the agent produce? And do the features that make an instrument usable by a robot survive from the real device to the admitted USD?

Figure 2 answers the first question with four stations that share nothing with Figure 1 except the compiler. Two are room-scale eBench packages in different laboratories, a teaching room and an analytical-instrumentation room, with a Lift2 already at the bench: a stoppered flask to open, and a glass rod to seat in a rack. Two are bench-level liquid stations: a stir bar to drop into a filled beaker, and a test tube to heat in a beaker water bath on a hotplate. With the Fehling bath, powder weighing, titration, and drying oven of Figure 1, the agent’s deliveries cover opening, unloading, weighing, racking, inserting, immersing, and turning a valve.

Figure 3 answers the second question on two instruments the agent compiled end to end. For the OVEN 125, the catalog photograph shows a right-hinged door swung open onto wire shelves; the specification writes that down as a revolute `door_hinge`, a handle collider, and shelves declared as

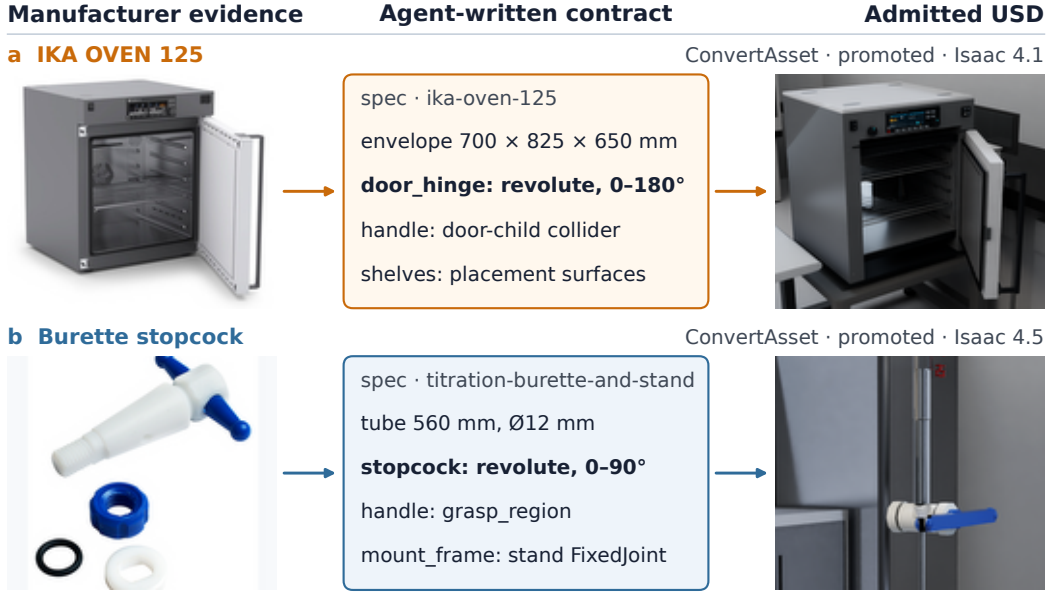


Figure 3: Two end-to-end agent traces read left to right. Manufacturer evidence is what the agent searched out; the contract is four fields from the specification it wrote; the admitted USD is the scene Isaac renders after ConvertAsset promotion (the oven asset under Isaac Sim 4.1, the titration station under 4.5). (a) IKA OVEN 125: the door hinge and shelves carry from the catalog view to the loaded, open oven. (b) Burette stopcock: the manufacturer’s PTFE stopcock becomes a revolute joint with a grasp handle on the graduated tube, mounted on the stand through a fixed joint.

placement surfaces; the admitted USD, rendered from the matching three-quarter view, opens the same way with a beaker on the shelf the specification named. For the burette, the manufacturer’s exploded stopcock becomes a revolute stopcock joint with a grasp region on its handle, and in the admitted scene that joint is the part a Lift2 turns. In both rows the middle column is what made the right column checkable.

5 CONCLUSION

A protocol and a USD scene used to be different documents written by different people. Lab2Sim is an agent that compiles one into the other: it searches out the instrument a protocol names, writes the specification that fixes what a robot must be able to grasp, open, and place, generates the articulated USD, and lets ConvertAsset decide whether the result may enter the laboratory. The stations in this paper are the compiler’s output, and the two traces in Figure 3 are the compiler’s audit trail. The next step is the obvious one: put the Lift2 to work in them.

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A PIPELINE DETAILS

Lab2Sim is implemented in Scenario Forge as a producer-side agent in front of the existing package compiler. Request understanding, search, specification, and generation are the new front-end. ConvertAsset, source bindings, asset locks, and scenario compile are unchanged.

Request. The agent reads the experiment requirement and emits an asset request list: one row per object, with a slug, why the experiment needs it, whether it articulates, which dimensions are task-critical, and the intended robot contact (grasp, pour, open, place). Text, stills, and video are parsed into that same table.

Evidence. For each row the agent retrieves manufacturer identity. Catalog pages, datasheets, and official photographs are hashed into an evidence register. Items are graded as exact-model evidence, family evidence, derived crops, or unknown. Unknown fields stay unknown; they are not filled in from pixels.

Specification. The specification template has a manufacturer table, a decomposition decision, a prim tree, joint contracts, state variables, interaction semantics, a QA matrix, delivery conventions, and an open-measurement punch list. Table 1 is the manufacturer table of the OVEN 125 specification behind Figure 3a. The burette specification fixes a revolute stopcock on `burette.usd`, a static `burette_stand.usd`, and a scene-level fixed joint between their mount frames.

Generation. The skill iterates geometry and articulation against the specification. Blender is the always-on modeller. Hunyuan3D is called only when the service is up; a miss falls back to Blender without changing the specification. A static articulation checker runs under Isaac Kit Python. Failures block admission.

Admission. ConvertAsset snapshots the generated tree, converts meshes and materials, and qualifies PhysX. A promoted asset carries a promotion receipt with its hash and the qualifying runtime; the scene that uses it carries a render manifest with camera poses and frame hashes. Figure 3 is built from those two records for each row: the OVEN 125 asset was promoted under Isaac Sim 4.1 with a 4.5 compatibility check, and the burette station under Isaac Sim 4.5. Lab2Sim does not own a second physics stack.

B OVEN 125 SPECIFICATION EXCERPT

Table 1: Manufacturer fields for the IKA OVEN 125 control-dry (solid door) as fixed in the agent-written asset specification.

Field	Value	Role
Envelope $W \times H \times D$	700×825×650 mm	Body collider
Chamber $W \times D \times H$	550×525×450 mm	Interior volume
Volume	125 L	Identity
Mass	82 kg	Fixed base
Door travel	~180° revolute	Open / close
Hinge	Configurable side	Joint axis
Shelves	Up to 6; 30 kg each	Placement surfaces
Controls	Dual TFT, rotary/push knob	Setpoint

Decomposition follows the specification rather than the mesh. `oven_body` is a fixed base. `door` is a revolute link. `handle` is a grasp collider on the door. `shelf_0–shelf_5` carry placement surfaces. `control_knob` is a rotary/push compound. Generation may rebuild geometry. It may not rename those contracts.

C ADDITIONAL RENDERS

Figure 4 adds three close views. The oven interior shows the lower shelf of the admitted OVEN 125 (r2), with a beaker and flask seated on the wire deck. The powder close-up is the weighing scoop of the r4 station in Figure 1. The Lift2 frame is the same Isaac runtime that consumes the compiled laboratories: a bimanual robot already spawned in front of a table.



Figure 4: Extra Isaac frames. Left: OVEN 125 lower shelf with glassware. Center: powder-weighing scoop. Right: Lift2 in the Isaac runtime that loads Lab2Sim scenes.

D FIGURE SOURCES

Figures 1–3 use frames from the current-delivery scenes and from the indexed manufacturer evidence. Figure 2 shows eBench task05 (r11), task07 (r10.1), the stir-bar station (r5), and the water-bath station (r1). Figure 3 uses the OVEN 125 Task 12 scene (r2) and the titration station (r1.7); the open-oven frame is a camera-only retake of the admitted scene chosen to match the catalog viewpoint. Frames are cropped and, where noted in the figure provenance, globally brightness-adjusted; scene state, geometry, materials, and physics are never edited for a figure. Every scene in the paper can be generated with the Blender skill alone.